

ReDNA Project Report

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SVM

Test Metrics:

Test Accuracy: 0.5720952380952381

Test Accuracy: 0.5720952380952381

Exact Match (EM): 0.5720952380952381

Top-3 Accuracy: 0.966

AUC: 0.8142897777777777

Precision/Recall/F1:

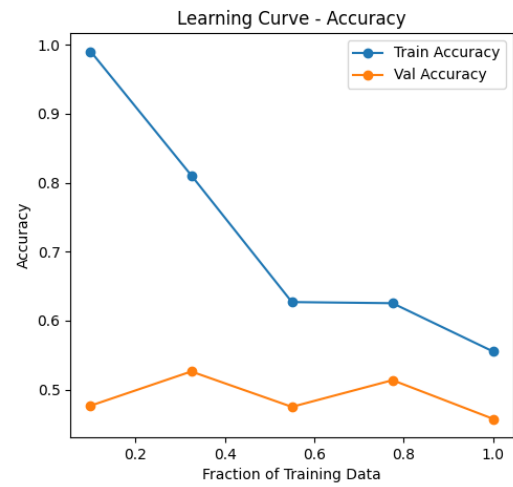
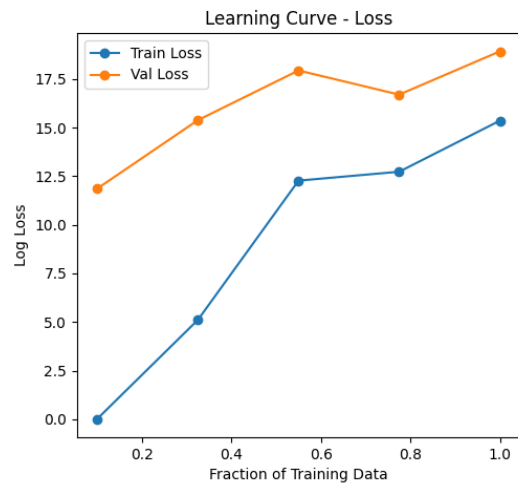
Class	Precision	Recall	F1-score	Support
cds	0.536	0.616	0.573	2625
intergenic	0.616	0.776	0.687	2625
promoter	0.295	0.175	0.220	2625
terminator	0.723	0.722	0.722	2625
macro avg	0.543	0.572	0.551	10500
weighted avg	0.543	0.572	0.551	10500

Confusion Matrix:

```
[[1616 264 650 95]
 [ 151 2037 209 228]
 [1179 585 460 401]
 [ 69 423 239 1894]]
```

Classes: ['cds', 'intergenic', 'promoter', 'terminator']

Performance Graph:



Naive Bayes

Test Metrics:

Test Accuracy: 0.4775238095238095

Exact Match (EM): 0.4775238095238095

Top-3 Accuracy: 0.929047619047619

AUC: 0.7401600483749055

Precision/Recall/F1:

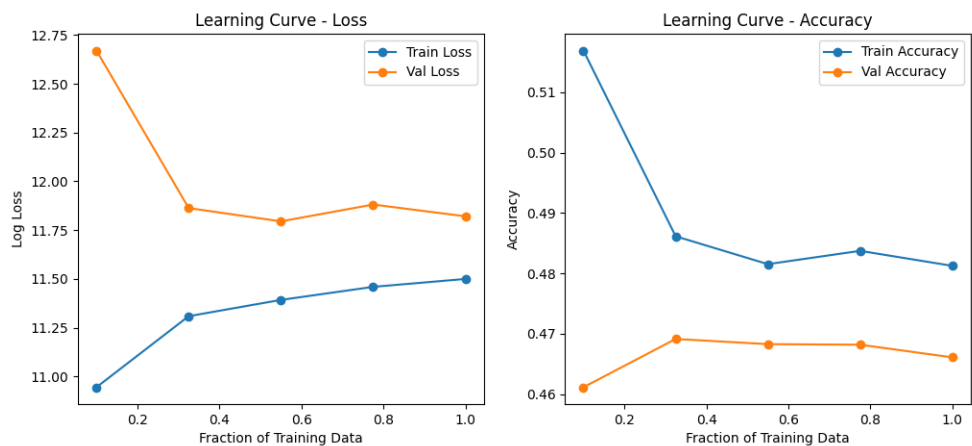
Class	Precision	Recall	F1-score	Support
cds	0.447	0.689	0.542	2625
intergenic	0.579	0.322	0.413	2625
promoter	0.275	0.140	0.186	2625
terminator	0.545	0.760	0.634	2625
macro avg	0.461	0.478	0.444	10500
weighted avg	0.461	0.478	0.444	10500

Confusion Matrix:

```
[[1808  88 452 277]
 [ 811 844 199 771]
 [1369 269 368 619]
 [  56 257 318 1994]]
```

Classes: ['cds', 'intergenic', 'promoter', 'terminator']

Performance Graph:



RNN

Test Metrics:

Test Accuracy: 0.364952380952381

Exact Match (EM): 0.364952380952381

Top-3 Accuracy: 0.8505714285714285

AUC: 0.6437933424036281

Precision/Recall/F1:

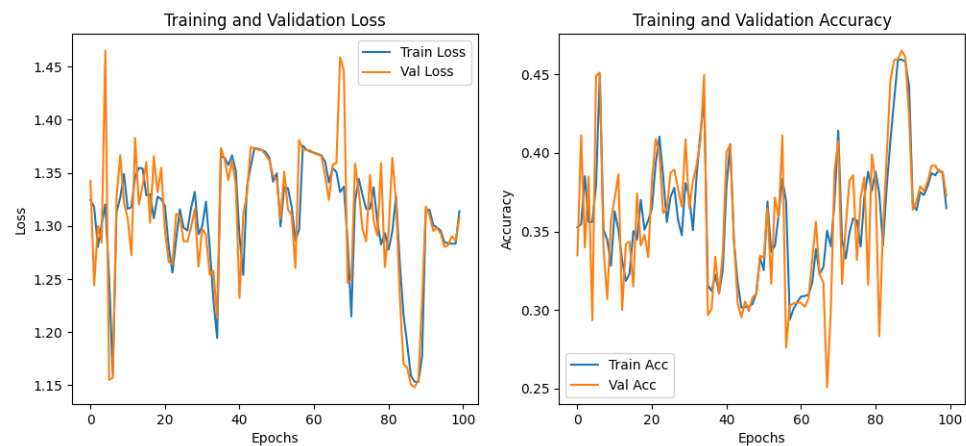
Class	Precision	Recall	F1-score	Support
cds	0.400	0.470	0.432	2625
intergenic	0.328	0.296	0.311	2625
promoter	0.279	0.079	0.123	2625
terminator	0.375	0.615	0.466	2625
macro avg	0.346	0.365	0.333	10500
weighted avg	0.346	0.365	0.333	10500

Confusion Matrix:

```
[[1233 464 238 690]
 [ 554 778 148 1145]
 [ 983 584 207 851]
 [ 316 547 148 1614]]
```

Classes: ['cds', 'intergenic', 'promoter', 'terminator']

Performance Graph:



Bi-LSTM

Test Metrics:

Test Accuracy: 0.7212380952380952

Exact Match (EM): 0.7212380952380952

Top-3 Accuracy: 0.9980952380952381

AUC: 0.8264929765684051

Precision/Recall/F1:

Class	Precision	Recall	F1-score	Support
cds	0.294	0.012	0.023	2625
intergenic	0.588	0.966	0.731	2625
promoter	0.829	0.918	0.871	2625
terminator	0.819	0.989	0.896	2625
macro avg	0.632	0.721	0.630	10500
weighted avg	0.632	0.721	0.630	10500

Confusion Matrix:

```
[[ 32 1709 452 432]
 [ 54 2536 17 18]
 [ 23 69 2409 124]
 [ 0 0 29 2596]]
```

Classes: ['cds', 'intergenic', 'promoter', 'terminator']

Performance Graph:



Transformer

Test Metrics:

Test Accuracy: 0.7241904761904762

Exact Match (EM): 0.7241904761904762

Top-3 Accuracy: 0.9989523809523809

AUC: 0.8298257173091459

Precision/Recall/F1:

Class	Precision	Recall	F1-score	Support
cds	0.330	0.013	0.024	2625
intergenic	0.596	0.963	0.736	2625
promoter	0.817	0.993	0.897	2625
terminator	0.821	0.928	0.872	2625
macro avg	0.641	0.724	0.632	10500
weighted avg	0.641	0.724	0.632	10500

Confusion Matrix:

```
[[ 33 1680 433 479]
 [ 48 2527 11 39]
 [ 4 22607 12]
 [ 15 34 139 2437]]
```

Classes: ['cds', 'intergenic', 'promoter', 'terminator']

Performance Graph:

